Early and Explainable Student Performance Prediction

Open University Learning Analytics Dataset (OULAD)

Report on how the code works

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1.Aim of the project

The notebook builds an **early warning system** for students of the Open University (about 32,000 course enrolments). Using only the **first 28 days** of a course, it predicts whether a student will end as **High Risk** (Fail or Withdrawn), **Medium Risk** (Pass) or **Low Risk** (Distinction), explains *why*, and suggests an *action*.

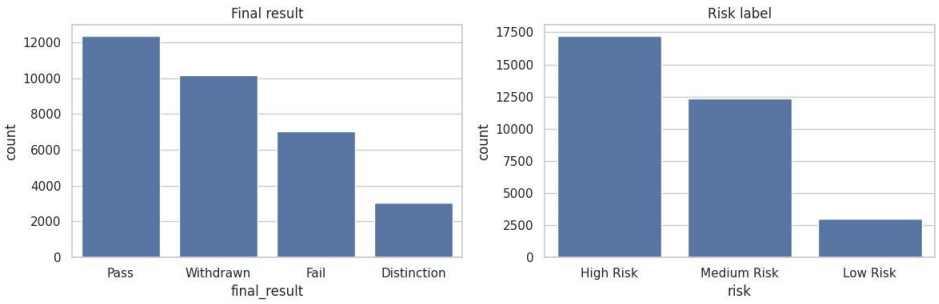
Flow: Early data → Machine learning → Predict risk → Explain why → Suggest intervention.

2.Dataset

Six OULAD tables are used. Dates are in days relative to the course start (day 0).

Table	Rows	Content
studentInfo	32,593	Demographics, previous attempts, credits, final result
studentRegistration	32,593	Registration and un-registration dates
assessments / studentAssessment	206 / 173,912	Assessment type, due date, weight / each submission and score
studentVle	10,655,280	Daily clicks of each student on the online platform
vle	6,364	Activity type of each platform page (forum, quiz, content...)

The final result is converted to the risk label: Fail/Withdrawn = High (17,208 students), Pass = Medium (12,361), Distinction = Low (3,024). The label is the answer to predict and is never used as an input.



Final result and risk label distribution.

3.Feature building

The function `build_features(t)` creates one row per student with **44 features from four groups**, using only events up to day t. Students who had already un-registered before day t are removed (5,055 at t = 28), leaving 27,538 students.

Group	Examples	No.
Student profile	education level, age band, deprivation band, previous attempts, credits	8
Course context	module, semester, registration date	3

Behavioural engagement	total clicks, active days, longest gap, days since last activity, clicks in last 7/14 days, clicks per activity type	20
Group	**Examples**	**No.**
Academic progress	assessments missed or late, submit rate, average / minimum / last score, weighted score	13

Leakage controls

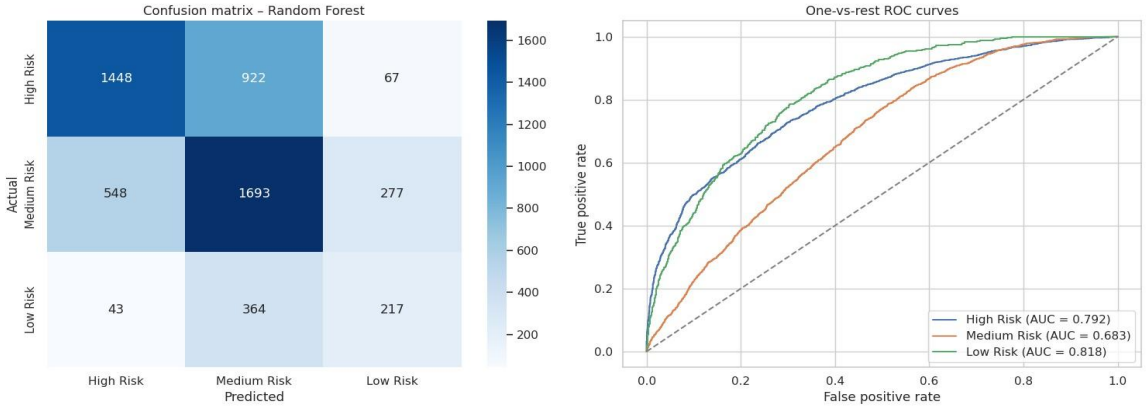
- Every feature uses only data with date <= t; exams are excluded because they happen at the end.
- A group split on student id keeps any student out of both train and test (21,959 train / 5,579 test).
- Model selection and threshold tuning use cross-validation on the training set only.

4.Models and evaluation

Preprocessing (inside a pipeline): median / most-frequent imputation, quantile scaling for numbers, one-hot encoding for categories. **Five models** are compared by 5-fold grouped cross-validation on macro-F1: Logistic Regression, Random Forest, Extra Trees, HistGradientBoosting and a Voting Ensemble. Random Forest is chosen (CV macro-F1 0.551).

Baseline: Mutual Information gives each feature a weight, and a simple Weighted Risk Index combines 13 indicators into one score (test accuracy 0.552). The ML models must beat this baseline.

Model (test set, day 28)	Accuracy	Macro F1	ROC-AUC
Random Forest (selected by CV)	0.602	0.543	0.746
Voting Ensemble	0.582	0.547	0.750
Extra Trees	0.577	0.537	0.742
HistGradientBoosting	0.558	0.528	0.746
Logistic Regression	0.539	0.513	0.735
Weighted Risk Index (baseline)	0.552	-	-



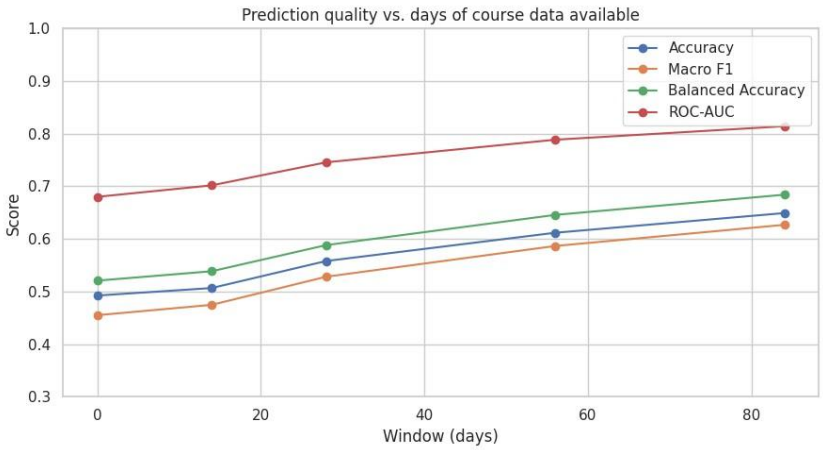
Confusion matrix and ROC curves of the Random Forest

Alert threshold. Missing an at-risk student is costly, so the High-Risk threshold is tuned on out-of-fold training predictions to reach at least 85% recall. The selected threshold is 0.26. On the test set, High-Risk recall rises from 59% to 87% while precision falls from 71% to 57%.

Setting	Precision	Recall	F1
Default (highest probability)	0.710	0.594	0.647
Tuned (threshold 0.26)	0.567	0.872	0.687

5.Extra analyses

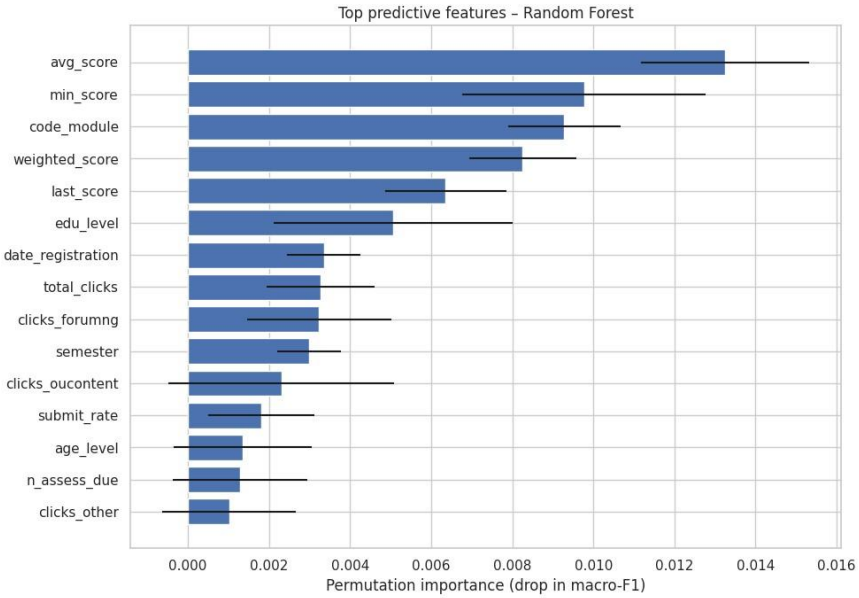
- **How early?** ROC-AUC is 0.68 at day 0, 0.70 at day 14, 0.75 at day 28, 0.79 at day 56 and 0.81 at day 84, so prediction improves as more data arrive.
- **Ablation:** each group alone is weaker (profile only ROC-AUC 0.60; engagement 0.68; academic 0.70); all four together give the best result (0.746).
- **Temporal test:** training on 2013 and testing on 2014 gives accuracy 0.53 and ROC-AUC 0.72, showing a realistic drop on unseen course runs.



Prediction quality versus days of data available

6.Explanation and intervention

Global explanation: permutation importance shuffles one feature at a time and measures the drop in macro-F1. The most important features are average score, minimum score, weighted score, last score, course module and education level.



Permutation importance of top features.

Per-student explanation: each actionable feature is replaced one at a time by the median value of students who did not fail. The drop in P(High Risk) is that feature's risk contribution. Gender, region, disability, age and deprivation are excluded from explanations and interventions because it would be unfair to target students by them.

Intervention rules: each top driver is mapped to a reason and an action, for example:

Driver	Recommended action
Recently inactive (days_since_last_active)	Immediate outreach - contact within 48 hours
Missed assessments / low submit rate	Contact the student; offer an extension plan
Low average score	Assign tutor support and targeted revision
Little forum activity	Invite to a study group or forum discussion
No pre-course preparation	Share the preparation checklist and orientation material

Example output for one student:

```
HIGH WARNING: immediate academic intervention recommended.
Probabilities: High 30.1% | Medium 52.0% | Low 17.8%
Why: Rarely visits course home (+2.2%), Shallow study sessions (+1.6%),
     No pre-course preparation (+1.3%) Action: Send weekly roadmap; suggest focused
study blocks; share preparation checklist.
```

Practical value: with the tuned threshold, 3,746 of 5,579 test students are flagged and 313 of 2,437 truly High-Risk students are missed. If tutors contact only the top 10% ranked students, 92.5% of them are truly High Risk.

7.Limitations and conclusion

- Three-class accuracy is moderate (about 60%); Low Risk (Distinction) is the hardest class (F1 about 0.37).

- Performance drops on later course runs, so the model needs periodic re-training.
- Explanations show model sensitivity, not proven causes, and the real effect of interventions was not tested.

Conclusion: the system can flag at-risk students within the first four weeks, explain the reasons and suggest support actions, and it improves as more course data arrive.